

YING ZHANG

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Statistics graduate (First Class), MPhil student in Data Intensive Science at Cambridge. First-author manuscript under review deriving closed-form layers for hyperbolic neural networks, one co-first-author manuscript under review, and two accepted papers on credit-risk time series.

EDUCATION

UNIVERSITY OF CAMBRIDGE

MPhil in Data Intensive Science, Lucy Cavendish College

Cambridge, UK

Oct 2026 - July 2027

UNIVERSITY OF NOTTINGHAM

Bachelor of Science (Honours), Statistics - First Class

Nottingham, UK

Oct 2022 - June 2026

Major Courses: Complex Analysis, Differential Equations, Vector Calculus and Electromagnetism, Real Analysis, Scientific Computation, Foundation of Pure Maths, Applied Mathematics, Analytical Computing, Statistical Inference (97), Probability and Statistics (90), Complex Analysis (89)

RESEARCH EXPERIENCE

Compressed Latent Diffusion Language Models (CLDLM)

Hangzhou, CN

Research Intern, ZJU-UIUC, co-author manuscript under review

Jul 2026 – Sep 2026

- Addressed length-compressed diffusion language modeling, where prior two-stage pipelines fix an autoencoder before training the generative model, by jointly training a compressor, a Flow Matching model, and a token decoder end to end so the compressed representation adapts to generation.
- Coupled JEPA-style joint-embedding prediction with Flow Matching by using the Flow Matching model itself as the predictor at the final noise step, so one network both predicts and denoises the compressed embedding during generation.
- Benchmarked against seven autoregressive, discrete-diffusion, and flow-based baselines including ELF on LM1B and OpenWebText with five-seed evaluation, achieving the lowest generation perplexity and highest throughput among all diffusion and flow models, 26.20 Gen-PPL on OpenWebText at 2.4x ELF's throughput (10,511 vs. 4,416 tokens/s).

Klein Neural Networks++: Closed-Form Layers on the Klein Model of Hyperbolic Space

Trento, Italy

Research Intern, University of Trento, First-author manuscript under review

Oct 2025 – April 2026

- Derived a closed-form Klein batch normalisation that replaces the iterative Fréchet mean required by prior gyrogroup BN with a single-pass Einstein midpoint, removing the inner optimisation loop at negligible accuracy cost.
- Developed the first systematic set of intrinsic Klein layers, spanning classification, fully connected, and normalisation heads in both geodesic and horosphere form, from a single Klein hyperplane distance. The model had stayed unexplored while the field standardised on the Poincaré ball and Lorentz hyperboloid, despite being the only one of the three admitting both Euclidean geodesics and a closed-form midpoint.
- Benchmarked against Poincaré, Lorentz, and PV counterparts on image classification (CIFAR-10/100, Tiny-ImageNet, ImageNet-1k), graph link prediction, and node classification, where Klein matches or improves on all three while remaining stable under feature clipping, curvature mis-specification, and gradient scaling.

When Early Is Not Early: The Precursor Illusion in Time-Series Anomaly Detection

Beijing, CN

Research Intern, Tsinghua University, Co-first-author manuscript under review

Apr 2026 – Aug 2026

- Formalised the precursor illusion where pre-onset labels are defined by temporal position alone and point adjustment is applied, causing the reactive detectors to appear to deliver advance warning they never had.
- Designed a model-agnostic, placebo-calibrated test for whether pre-onset intervals actually contain detectable evidence, finding fewer than one in seven benchmark events qualify once the placebo baseline is subtracted.
- Showed that after multiplicity correction none of 35 model-dataset cells across 7 detectors discriminates precursor from matched normal windows, while point adjustment inflates mean detection F1 from 0.186 to 0.549 with scores and thresholds held fixed.

Simulation Based Amortised Inference (SBI) using Neural Posterior Estimation ([github repo](#))

Nottingham, UK

Research Intern funded by Engineering and Physical Sciences Research Council (EPSRC)

Jun 2025 – Aug 2025

- Designed conditional normalizing flow architectures for simulation-based inference on i.i.d. Poisson models, recovering non-analytic posteriors in large-N regimes. Benchmarking against mixture-of-Gaussians density estimators showed the CLT-based regularisation improved stability over standard neural inference.
- Presented the work at the University of Nottingham Undergraduate Research Poster Exhibition (Nov 2025), covering the multi-distribution benchmarking and the CLT-based regularisation for an audience of faculty and students.

PUBLICATIONS AND MANUSCRIPTS

- **Ying Zhang**, Nicu Sebe, Bernhard Schölkopf, Ziheng Chen, *Klein Neural Networks++*, Under Review (first author)
- Yue Yang*, **Ying Zhang***, Yiwen Yan, Chen Zhang, *When Early Is Not Early: The Precursor Illusion in Time-Series Anomaly Detection*, Under Review (*equal contribution)
- Yulin Yuan, **Ying Zhang**, Xiangming Meng, *One Latent, Many Tokens: Jointly Learning Compressed Representations for Efficient Language Diffusion*, Under Review
- Yue Yang, Zihan Su, **Ying Zhang**, Chang Chuan Goh, Yuxiang Lin, Anthony Graham Bellotti, Boon Giin Lee, *Kolmogorov–Arnold Networks-based GRU and LSTM for Loan Default Early Prediction*, Applied Soft Computing 2025, Accepted (arXiv:[2507.13685](#))
- Yue Yang, Yuxiang Lin, **Ying Zhang**, Zihan Su, Chang Chuan Goh, Tangtangfang Fang, Anthony Graham Bellotti, Boon Giin Lee, *Transforming Credit Risk Analysis: A Time-Series-Driven ResE-BiLSTM Framework for Post-Loan Default Detection, The Third Credit Scoring and Credit Rating Conference (CSCR III)*, 2024, Accepted (arXiv:[2508.00415](#))

WORK EXPERIENCE

Trendy Group (Ochirly, Miss Sixty & Five Plus) ([page](#))

Guangzhou, CN

Data Analyst Intern

Jun – Aug 2024

- Designed, developed, and tested a rolling inventory system for Five Plus using IBM Planning Analytics, integrating master business data for automated calculations that improved product turnover.

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, NumPy, SciPy, scikit-learn, pandas, Jupyter, geoopt), R, LaTeX, Git
- **Methods:** Hyperbolic and Riemannian deep learning, Riemannian optimisation, normalizing flows, diffusion and score-based models, approximate and simulation-based inference, time-series anomaly detection
- **Research Computing:** Slurm job arrays and multi-GPU training on Ada (Nottingham) and CINECA Leonardo (EuroHPC), plus mixed-precision and numerical-stability debugging

HONORS AND AWARDS

- Asia and Pacific Mathematical Contest in Modeling (APMCM), First Prize, 2023 ([github repo](#), [paper](#))
- ShuWei Cup IMCM, Meritorious, 2023 ([github repo](#), [paper](#))
- China Undergraduate Mathematical Contest in Modeling (CUMCM), Zhejiang Provincial Third Prize, 2023 ([github repo](#), [paper](#))
- Huashu Cup National Mathematical Contest in Modeling, Third Prize, 2023 ([github repo](#), [paper](#))
- Mathematical Contest in Modeling (MCM), Successful Participant, 2024 ([github repo](#), [paper](#))
- Mathematical Contest in Modeling (MCM), Honorable Mention, 2025 ([github repo](#), [paper](#))